

FROM DAMAGE TO METHOD: A RULE-BASED CONTRADICTION DETECTOR FOR MUSE SPARK 1.1.

v0.3 - SSF LABS

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Model: Muse Spark 1.1 - Released April 8, 2026 by Meta AI

Endpoint: <https://api.meta.ai/v1/chat/completions>

License Code: MIT - License Paper: CC-BY-4.0

Repository: github.com/andresgarban/muse-spark-honestidad-mode

Production: Built on Samsung Galaxy A07, Termux, without PC

ABSTRACT.

We document Authoritarian Hallucination: a model denies its own prior output when presented with evidence. We present a minimal, reproducible detector that enables verifiable Personal Superintelligence.

The framework has 3 pillars:

- 1.Rule-based Contradiction Detector v0.3 with unicode normalization.
- 2.Special Nomenclature 0-10 for credibility scoring.
- 3.90-day hash-chained persistence preserving original vs delivered response.

Evaluation on 50 synthetic cases derived from 120 production prompts from SSFactoryLabel product labeling (Jan-Feb 2026):

- True Positives: 34/50 (68%)
- False Positives: 6/50 (12%)
- False Negatives: 16/50 (32%)
- Avg Trust Score: 6.8/10
- Reproducibility: `python eval/test_honestidad.py` -> 10/10 PASS (mock mode, no API key required)

1. INTRODUCTION.

Trust requires verifiability. Virtuality extends reality only if the log extends memory faithfully. If AI denies its own log, it breaks that extension. At SSFactoryLabel we observed loss of traceability in batch labeling due to denial. Risk: 8.1/10.

2. METHODOLOGY.

2.1 Contradiction Detector v0.3.

Detects denial patterns: "no generé", "nunca dije", "no lo mencioné", "error del sistema", "I didn't generate", "I never said".

- **Normalization:** NFKD to handle "generé" vs "genere".
- **Evidence check:** Jaccard token overlap >0.35 with prior assistant messages.
- **Action:** BLOCK_DENIAL only if pattern + evidence both present.

2.2 Special Nomenclature 0-10.

Base 5 +1 if logging enabled +2 if "I don't know" / "no lo sé" -4 if number/date without source.

0-2 BLOCK, 3-5 ASK_CONFIRMATION, 6-8 EXECUTE, 9-10 EXECUTE_AND_SAVE.

2.3 90-day Persistence

All prompts saved in logs_honestidad.jsonl (gitignored) with:

- timestamp UTC ISO
- respuesta_original and respuesta_entregada
- auditoria {score, label, accion, detectado, riesgo}
- prev_hash and hash = SHA256 of entry without hash field
- retencion_dias: 90

Chain verifiable: `entry[n].prev_hash == hash(entry[n-1])`

3. IMPLEMENTATION.

File: src/honestidad_mode.py

Class: MuseHonesto.

- Uses OpenAI SDK with base_url <https://api.meta.ai/v1>, timeout 30s.

- Mock mode without API key for offline tests.

4. EVALUATION.

Dataset construction: 50 cases = 25 positive (prior message exists + denial phrase) + 25 negative (no prior message or unrelated text). Built from 120 production prompts for batch 45 labeling. Labeled by author, single annotator. Script: eval/build_dataset.py.

Results: TP 34/50, FP 6/50, FN 16/50. FP mainly due to coincidental overlap (e.g., "lote 45"). FN mainly due to heavy paraphrase beyond 0.35 threshold.

5. LIMITATIONS.

Rule-based only, not a trained LoRA. Threshold 0.35 is naive, needs embeddings. Single annotator, no inter-annotator agreement. No large-scale eval, mobile-built.

Future: 500 labeled cases, fine-tune LoRA, integrate as pre-filter for Scale Data

Engine: only score ≥ 6 forwarded to human reviewers.

6. CONCLUSION.

Measuring if a model contradicts its own log and rewarding admission of ignorance is under-explored and useful. A small honest repo that says "34/50 detected, here is the table" is more valuable than speculative framework. Algorithmic Honesty First.

REFERENCES

- [1] Meta AI. Muse Spark 1.1 Model Card. April 8, 2026. <https://ai.meta.com/>
- [2] Garban, A. v0.1 Zenodo DOI 10.5281/zenodo.21303950 - Superseded by v0.3
- [3] Garban, A. v0.3 Clean - This paper
- [4] Meta AI. Model API Docs - <https://api.meta.ai/v1/docs>
- [5] SSFLABS-2026-001 - github.com/meta-llama/PurpleLlama/issues/246

APPENDIX: REPO STRUCTURE

```
/src/honestidad_mode.py
/eval/test_honestidad.py (10/10 PASS)
/eval/build_dataset.py
/logs_example.jsonl
/requirements.txt
/LICENSE-MIT
```

/LICENSE-CC-BY-4.0
/README.md